

Time Series

Basic Statistical Concepts

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Session Objectives

By the end of this session, you will be able to:

- Understand the fundamental statistical concepts used in time series analysis
- Define and interpret the moments of a distribution (mean, variance, skewness, kurtosis)
- Understand the main estimation methods:
 - Method of Moments (MOM)
 - Maximum Likelihood Estimation (MLE)
- Compute and interpret classical statistical estimators
- Understand the concept of sampling distributions
- Introduce the basic concepts of statistical inference:
 - Hypothesis testing
 - Confidence intervals

Basic Statistical Concepts

- Moments of a Distribution: Fully characterizing the distribution
- Estimation Methods: Method of Moments versus Maximum Likelihood Estimation
- Basic Estimators: Approach and Sampling Distribution
- Multivariate Data: Joint, marginal, and conditional distribution
- Statistical Inference: Confidence intervals and Hypothesis Testing

Moments of a Distribution

Moments of a random variable X with density $f(x)$:

- l -th moment:

$$m'_l = E[X^l] = \int_{-\infty}^{\infty} x^l f(x) dx$$

- l -th central moment:

$$m_l = E[(X - \mu)^l] = \int_{-\infty}^{\infty} (x - \mu)^l f(x) dx$$

- Examples:

- Expectation: $E[X]$
- Variance: $E[(X - \mu)^2]$
- Skewness: $S(X) = E\left[\frac{(X - \mu)^3}{\sigma^3}\right]$
- Kurtosis: $K(X) = E\left[\frac{(X - \mu)^4}{\sigma^4}\right]$

Statistical Estimation

Parametric Statistics:

Observe $x_1, \dots, x_n$ from random variables

$$X_1, \dots, X_n \sim f(x; \theta)$$

where θ is unknown.

Goal: Estimate θ using observations.

Approaches:

- Method of Moments (MOM)
- Maximum Likelihood Estimation (MLE)

Examples of Classic Estimators

Given data $\{X_1, \dots, X_n\}$:

- Sample mean:

$$\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

- Sample variance:

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{\mu})^2$$

Examples of Classic Estimators 2

Given data $\{X_1, \dots, X_n\}$:

- Sample skewness:

$$\hat{S} = \frac{1}{(n-1)\hat{\sigma}^3} \sum_{i=1}^n (X_i - \hat{\mu})^3$$

- Sample kurtosis:

$$\hat{K} = \frac{1}{(n-1)\hat{\sigma}^4} \sum_{i=1}^n (X_i - \hat{\mu})^4$$

Sampling Distribution of Estimators

Under normality assumption:

$$X_i \sim \mathcal{N}(\mu, \sigma^2)$$

$$\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$$

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

Properties:

- Unbiasedness
- Consistency

Method of Moments Estimation

Principle:

$$E[X^p] = \frac{1}{n} \sum_{i=1}^n X_i^p$$

Example:

$$X_i \sim \mathcal{N}(\mu, \sigma^2)$$

Mean estimator:

$$\hat{\mu} = \frac{1}{n} \sum X_i$$

Variance estimator:

$$\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$$

Maximum Likelihood Estimation

Likelihood function:

$$L(\theta) = f(x_1, \dots, x_n; \theta)$$

Under independence:

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta)$$

MLE estimator:

$$\hat{\theta} = \arg \max_{\theta} L(\theta)$$

Multivariate Distribution

Joint distribution:

$$f(x, y) = f(x|y)f(y)$$

For n variables:

$$f(x_1, \dots, x_n) = \prod_{i=1}^n f(x_i | x_{i-1}, \dots, x_1)$$

Example normal:

$$f(x) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(x_i - \mu_i)^2}{2\sigma_i^2}\right)$$

Hypothesis testing:

$$H_0 : \theta = \theta_0 \quad \text{vs} \quad H_a : \theta \neq \theta_0$$

P-value:

Measure of plausibility of H_0

Significance level:

Probability of type I error

Confidence interval:

$$\hat{\theta} \pm k$$

such that

$$P(\theta \in [\hat{\theta} - k, \hat{\theta} + k]) = 1 - \alpha$$